

BIOMECHANICS · EPHE 341

Forces

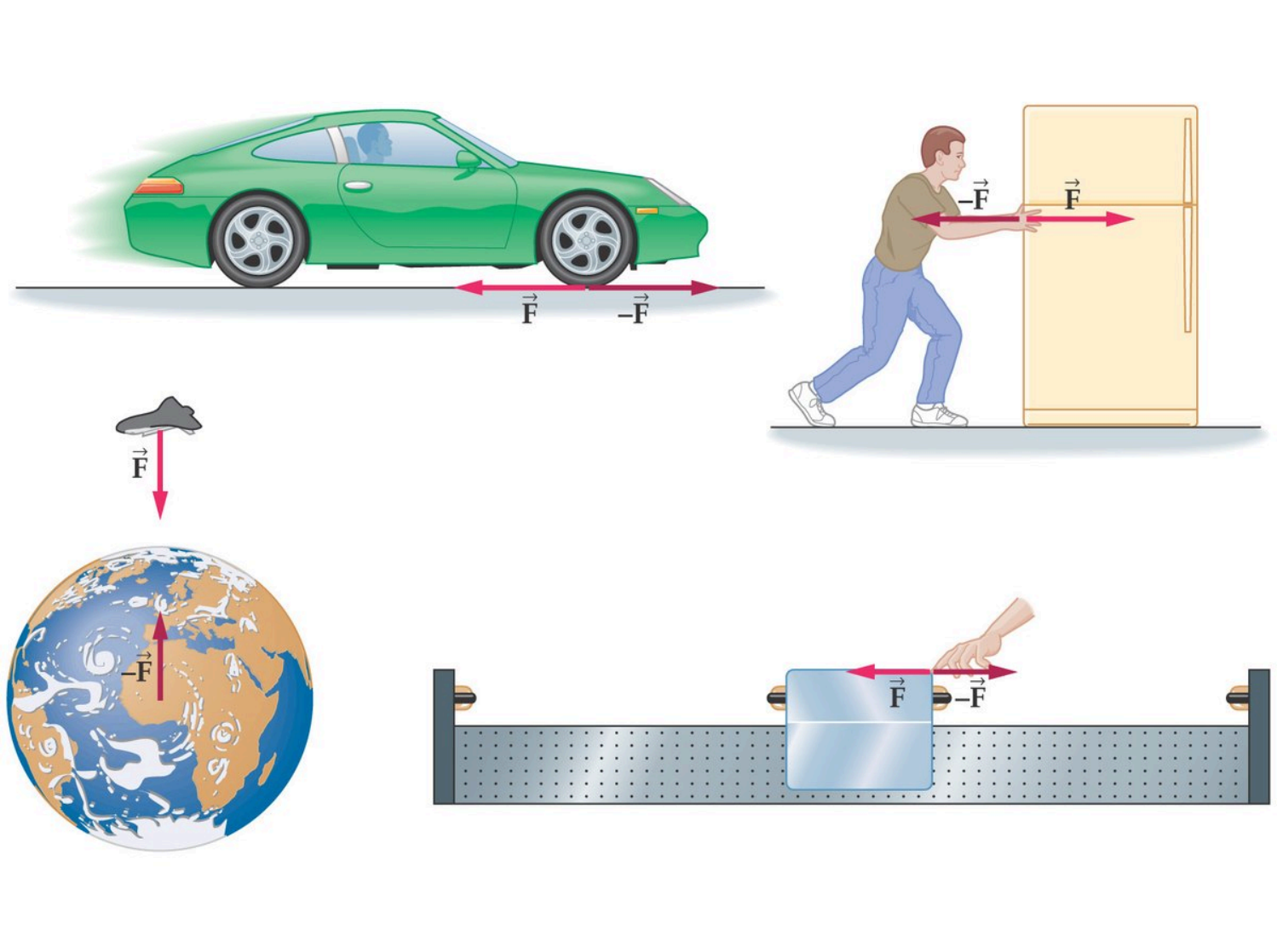
What a push or a pull actually is, how to add several of them together, and how we measure the ones we cannot see.

Force

Biomechanics is the study of the effects of **internal and external forces** on the human body in movement and rest.

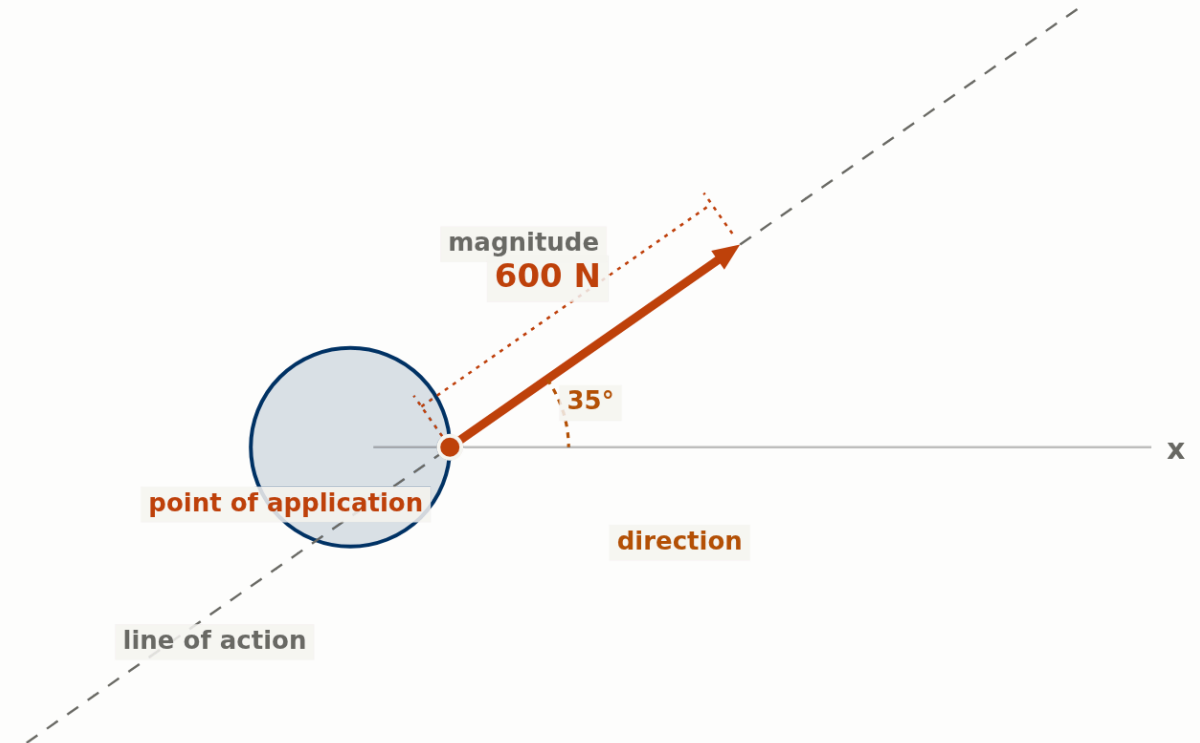
Simply defined, a force is a **push or a pull**.

- A force is a **vector** quantity.
- It can change an object's state of motion — start, stop, speed up, slow down, or change direction.
- It can **accelerate** and/or **deform** an object.
- It comes in pairs: action and reaction.



A force needs four things said about it before it is fully described. Drag the arrow and watch which of the four each control changes.

- **Magnitude** — the length of the arrow
- **Direction** — relative to an axis
- **Line of action** — the shaft of the arrow, extended both ways
- **Point of application** — where it acts on the object

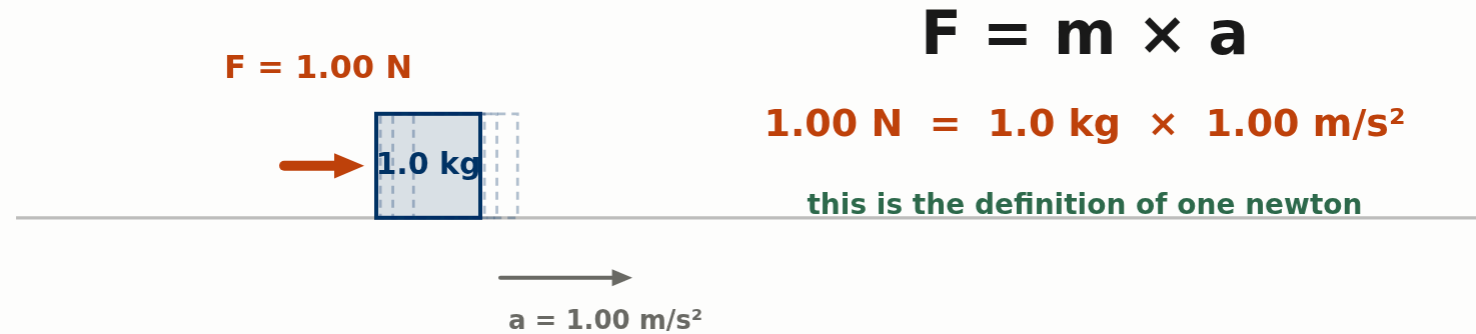


600 N at **35°** above the x axis · applied at the **centre**

Magnitude changes the length, direction swings the whole line of action, and the point of application decides where on the object it acts. Move it off the centre.

Force — units

The **newton (N)** is the SI unit of force. One newton is the force required to accelerate a 1 kg mass at 1 m/s².



$$1.00 \text{ N} = 1.0 \text{ kg} \times 1.00 \text{ m/s}^2$$

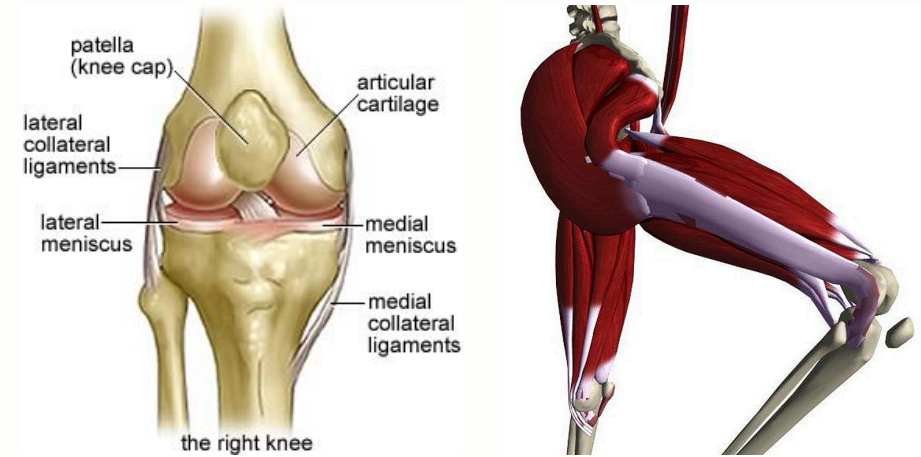
One newton is the force that accelerates one kilogram at one metre per second squared — set both sliders to 1 and the equation says so. Doubling either one doubles the force.

Mass	<input type="range"/>	1.0 kg
Acceleration	<input type="range"/>	1.00 m/s ²

Internal forces

Internal forces act **within** an object or system — **compressive** (pushing) and **tensile** (pulling).

When you land from a jump there is a force from your lower leg on your upper leg through your knee, and an equal and opposite force from your upper leg on your lower leg through the same joint.

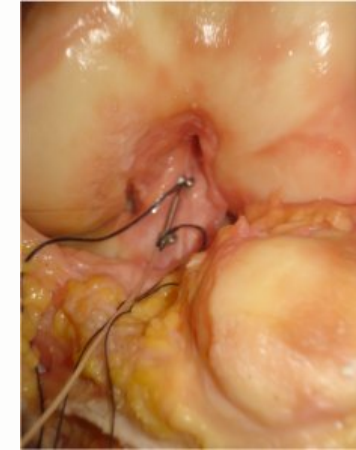
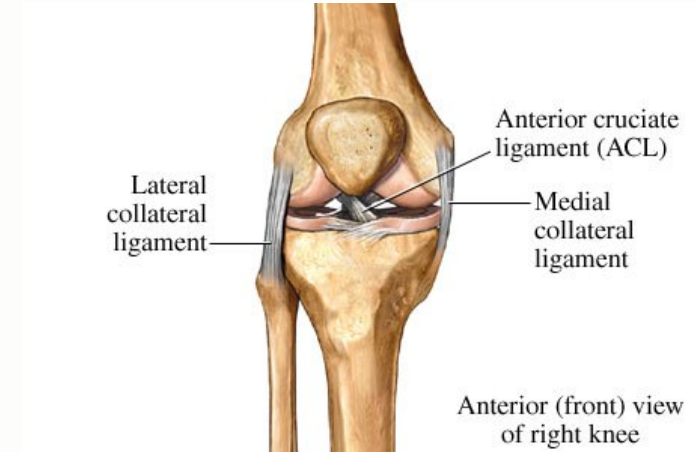


Measuring internal forces

It is difficult to measure internal forces in human biomechanics **directly**.

One invasive option is strain gauges in **cadaver limbs**, used to study how injuries happen and how they might be prevented.

The strain gauge is the same device as in lecture 2 — a resistance that changes when the material it is bonded to deforms.

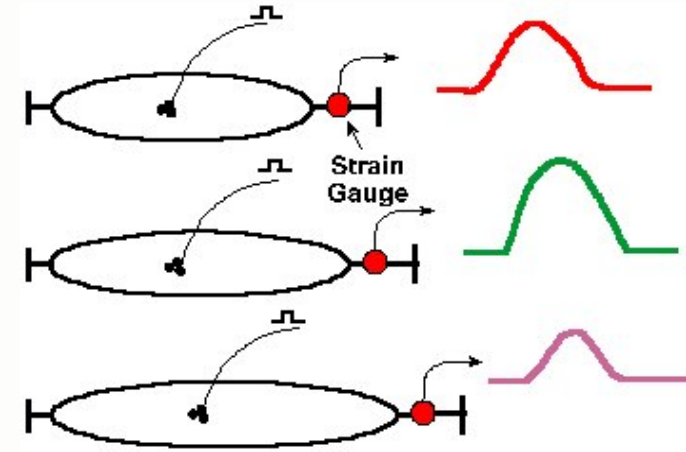


Measuring internal forces

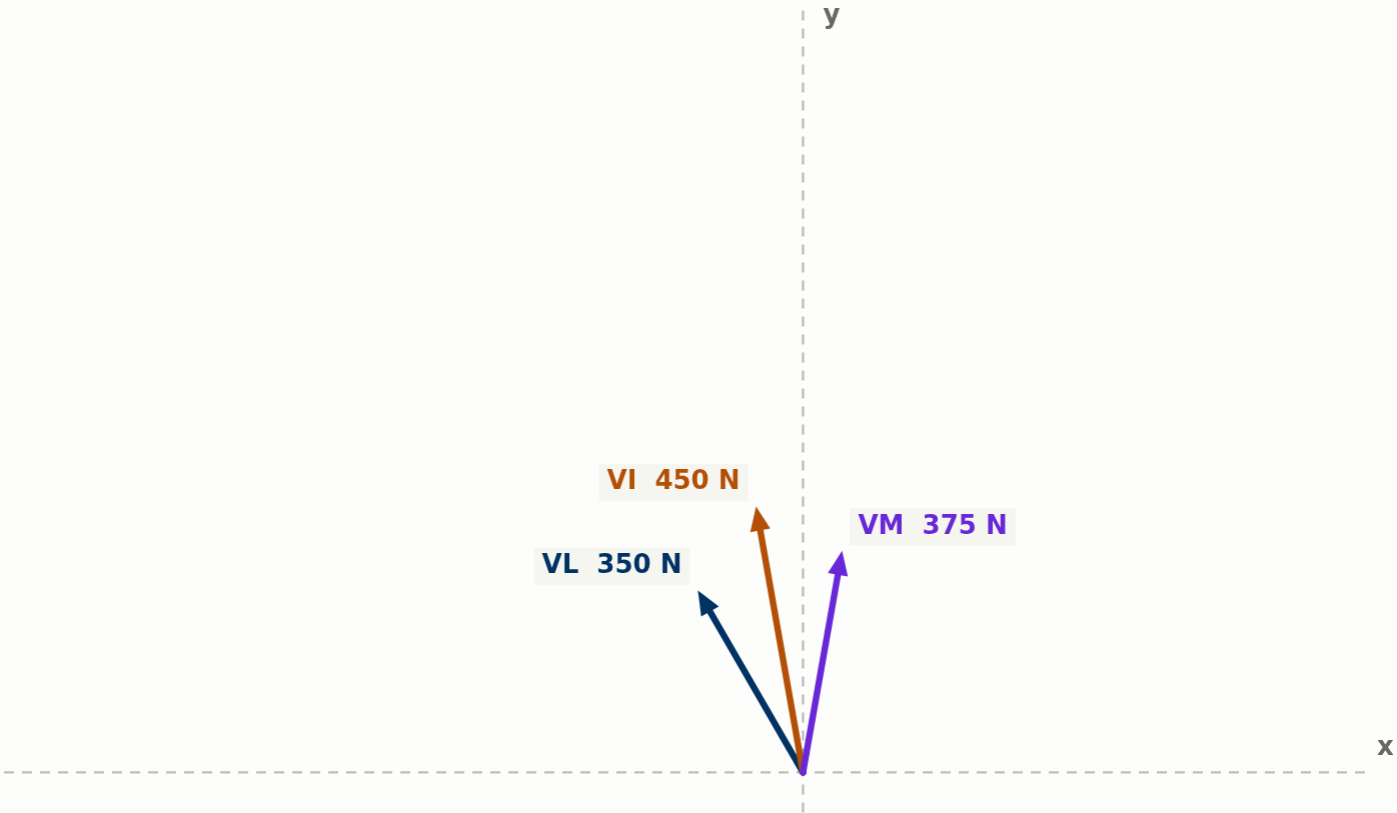
A strain gauge or optical fibre on or in a human or animal **tendon** can measure the **magnitude** of the muscle force.

But what about the **direction**?

A gauge tells you how hard the tendon is being pulled along its own length. It says nothing about where that pull points once the joint moves.



Fibre optic force measurement gives the individual quadriceps forces during a squat. **Find the resultant quadriceps force.**

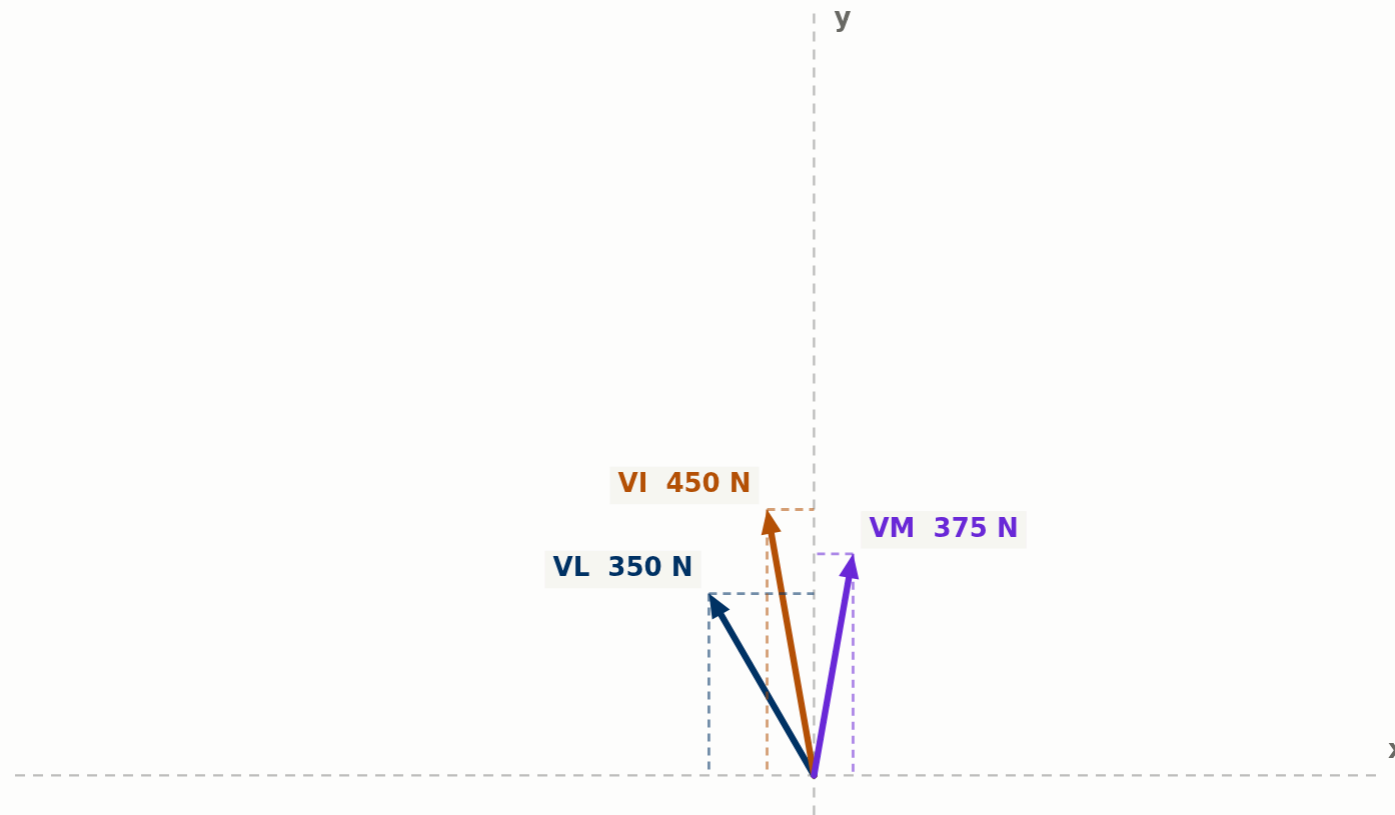


Step 2 · draw what is known

	force	angle
VL	350 N	60°
VI	450 N	80°
VM	375 N	80°

The angle is measured from the horizontal; VL and VI pull to one side of the midline, VM to the other.

Step 3. Isolate each force and find its components with the trig functions. Pick a muscle and watch its two components appear.



Step 3 · components of each force

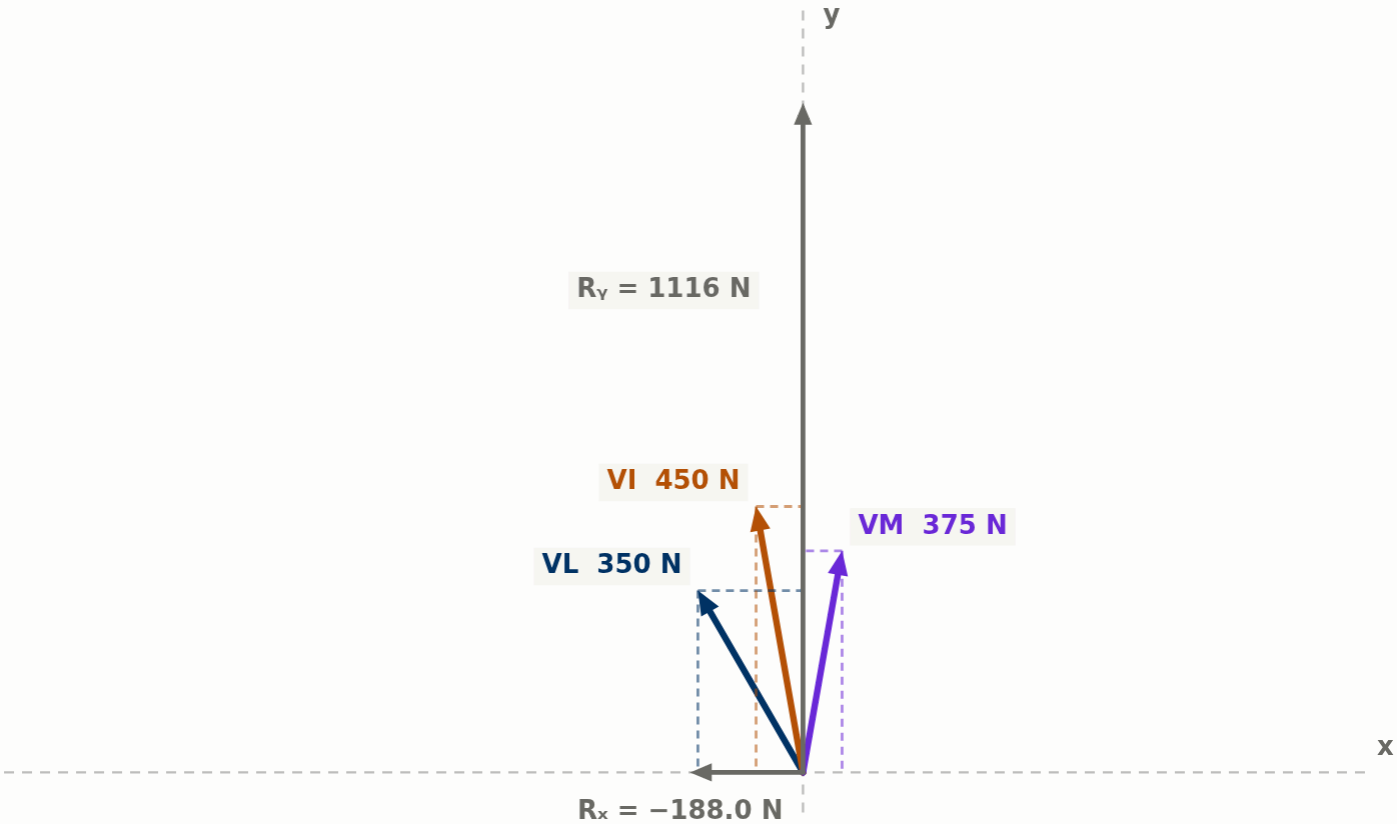
VL — vastus lateralis

$$F_x = F \cdot \cos\theta = 350 \cdot \cos 120^\circ = -175.0 \text{ N}$$

$$F_y = F \cdot \sin\theta = 350 \cdot \sin 120^\circ = 303.1 \text{ N}$$

	F_x	F_y
VL	-175.0	303.1
VI	-78.1	443.2
VM	65.1	369.3

Step 4. Add the components — horizontal with horizontal, vertical with vertical. The same table you used for linear kinematics.

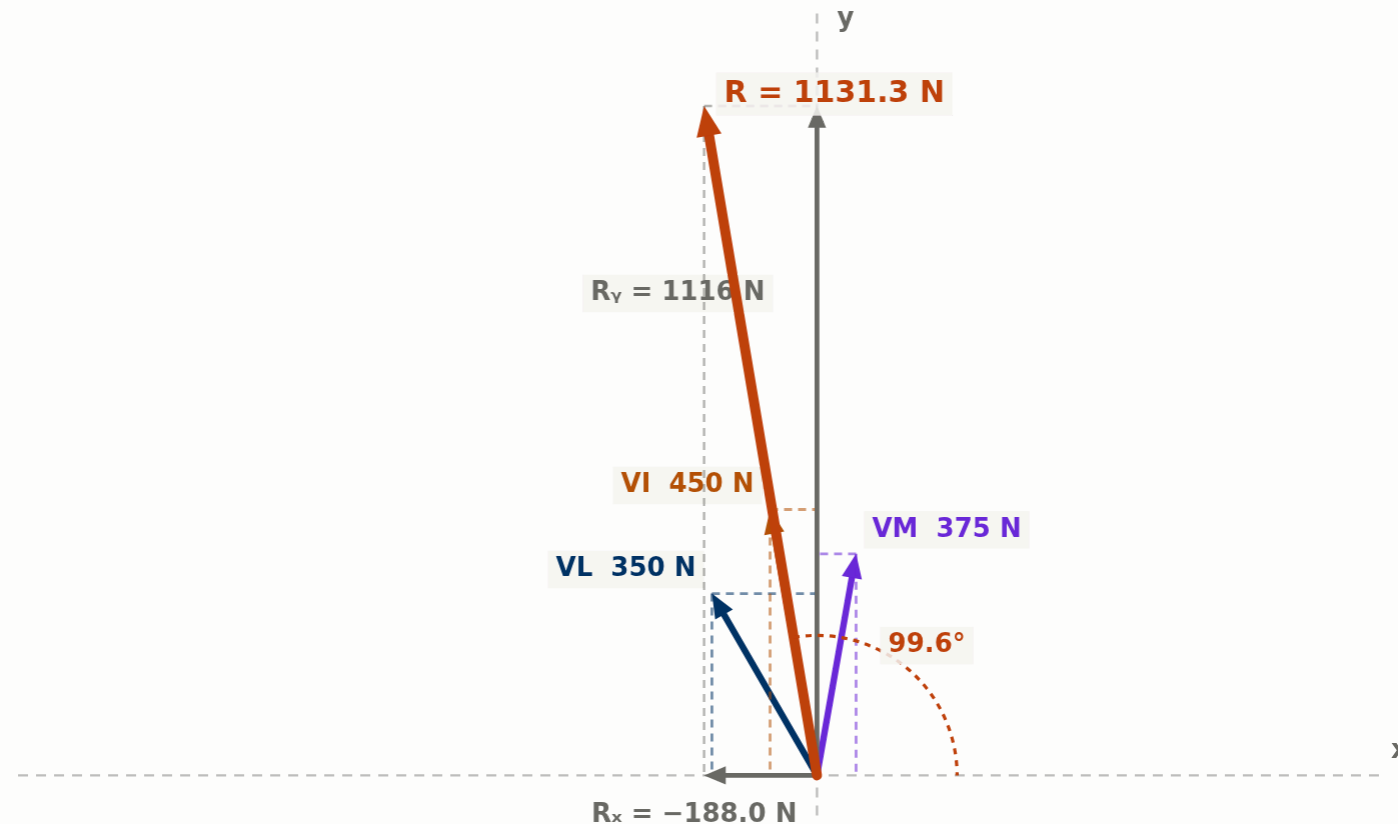


Step 4 · add the components

	F _x	F _y
VL	-175.0	303.1
VI	-78.1	443.2
VM	65.1	369.3
Σ	-188.0	1115.6

Horizontal with horizontal, vertical with vertical — the same table you used for kinematics.

Step 5. Pythagoras for the magnitude, the tangent for the direction — and then check which quadrant the answer is actually in.



Step 5 · magnitude and direction

magnitude

$$R^2 = R_x^2 + R_y^2$$

$$R = 1131.3 \text{ N}$$

direction

$$\tan\theta = R_y / R_x = 1116 / -188.0$$

$$\theta = 99.6^\circ \text{ from the x axis}$$

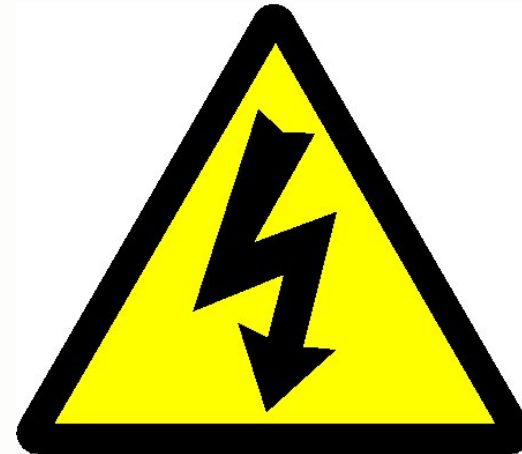
Watch the quadrant. A calculator given $1116 / -188.0$ returns -80.4° , which points down and to the right. R_x is negative and R_y is positive, so the resultant is up and to the left: add 180° to get 99.6° .

External forces

Forces that act on an object as a result of its **surroundings**. Two kinds:

- **Non-contact** — gravity, magnetic, electrical
- **Contact** — between objects touching each other, solid or fluid

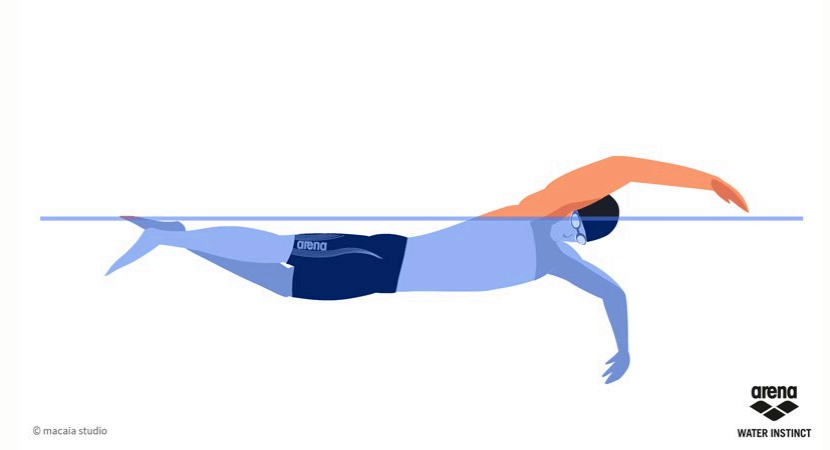
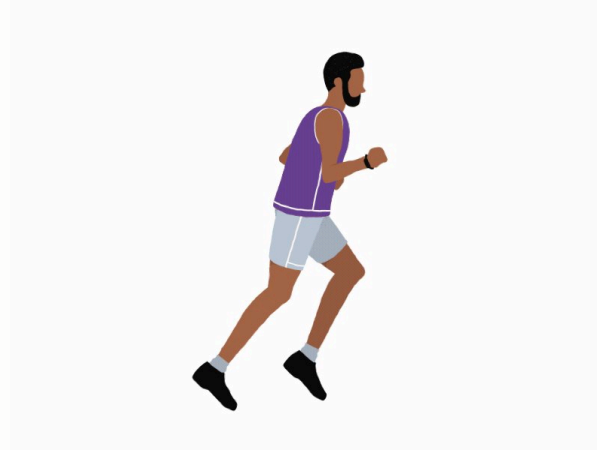
Air resistance and water resistance are fluid contact forces. Contact forces occur in pairs: action and reaction.



Gravity and contact forces

The force of gravity acting on us is what causes us to **contact the ground** — and the ground pushes back.

In the water or in the air, the contact force is spread over the whole surface of the body instead of the soles of the feet.



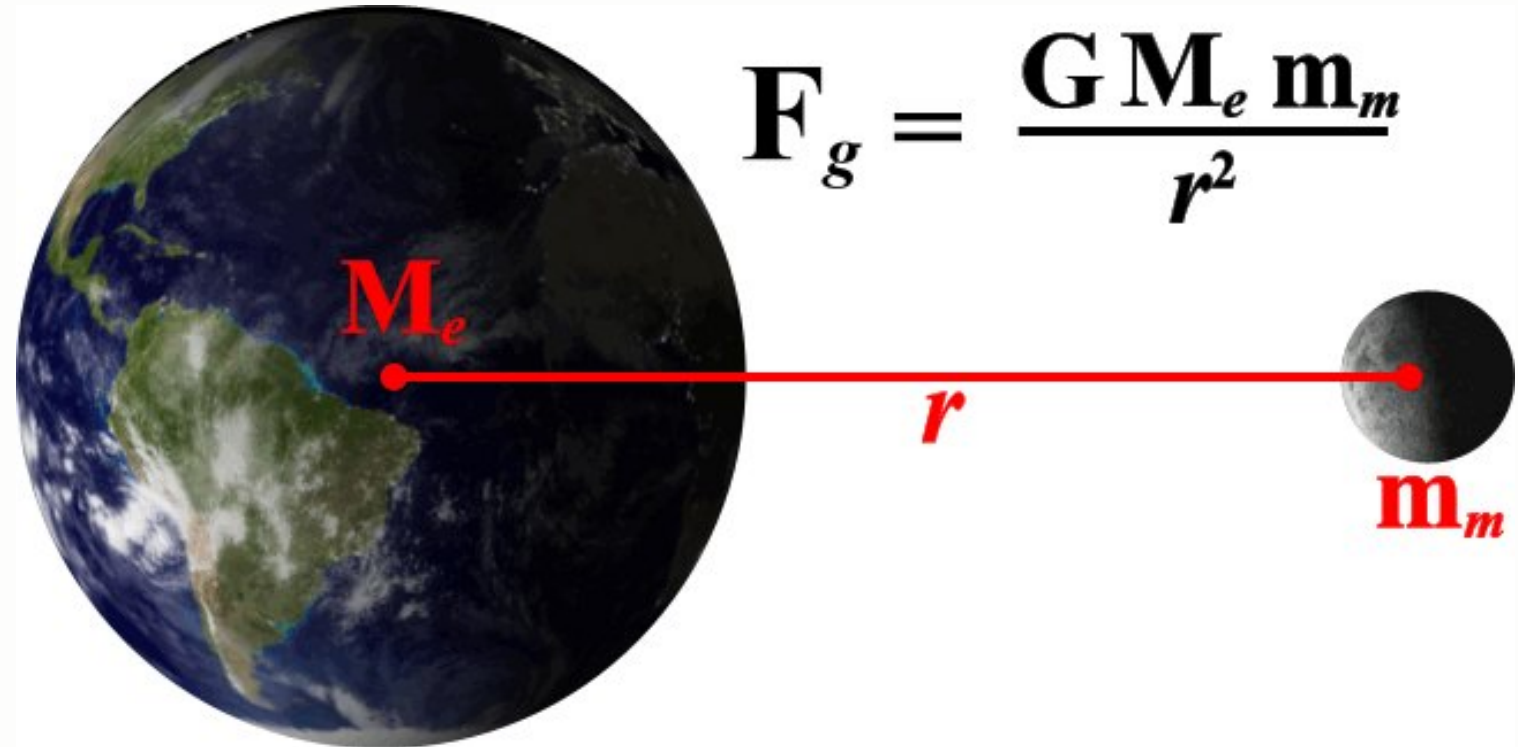
Gravity

We experience gravity as **weight**.

All objects with mass attract other objects with mass. The magnitude depends on both masses and the distance between them.

The Earth attracts you, and you attract the Earth with an equal and opposite force.

The result is a constant force on us, pointed at the centre of the Earth and acting through our **centre of mass**.



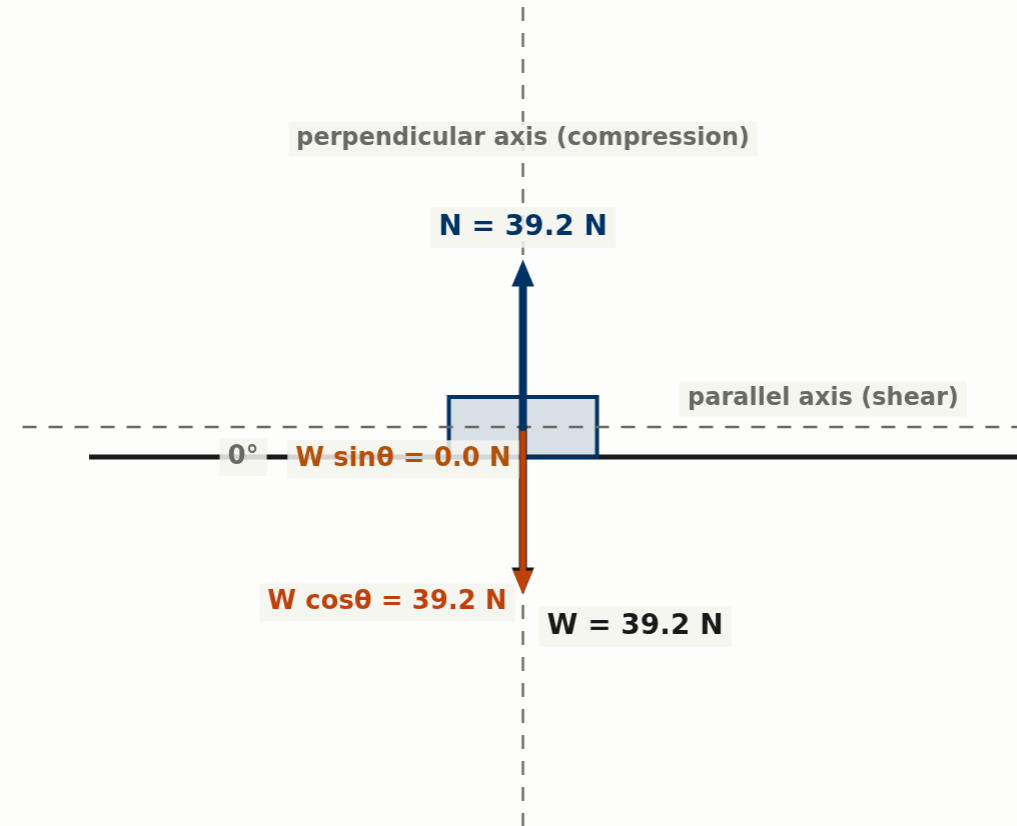
Weight is the **force** of gravity acting on an object: $W = mg$, with $g = 9.81 \text{ m/s}^2$. So — what is your weight in newtons?

Gravity and the normal force

Gravity acts on us towards the Earth, through our centre of mass. That causes us to contact the ground — and the ground must push back to hold us up.

The component of the ground's force acting **perpendicular to the contact surface** is called the **normal force**.

On a surface parallel to the horizontal, the normal force is **equal in magnitude and opposite in direction** to the force of gravity.



Flat ground: the normal force is **equal and opposite** to the weight, 39.2 N.

Gravity never changes — it is the same arrow straight down at every angle. What changes is the axis system we measure it against, which is why the normal force falls away as the slope steepens.

Slope angle  0°

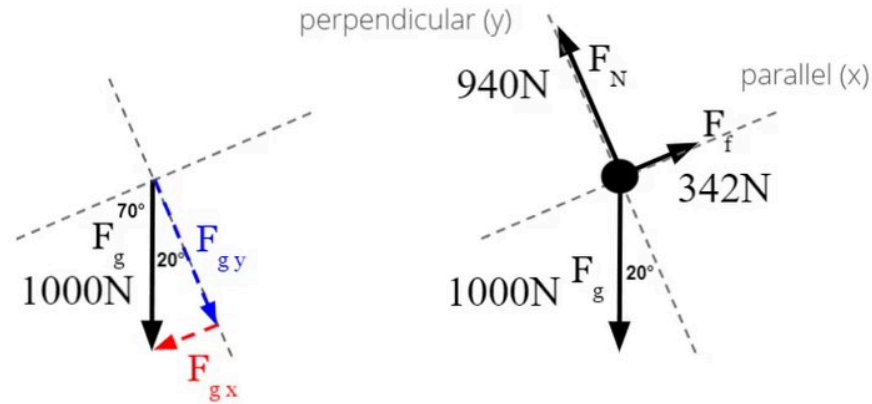
On a slanted surface

On a slope the force of gravity **still points straight at the centre of the Earth** — it does not tilt with the ground.

But the normal force stays **perpendicular to the contact surface**, so it is no longer equal and opposite to gravity.

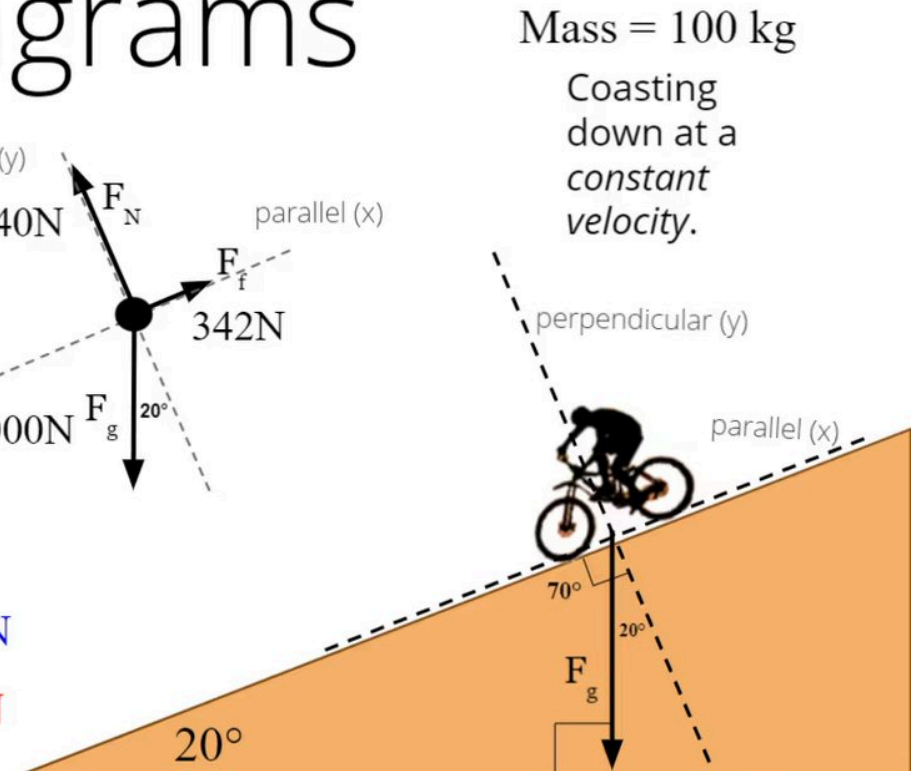
So how do we calculate the normal force?

Force Diagrams



$$F_{gy} = 1000\text{N} \cdot \cos(20^\circ) = 940\text{N}$$

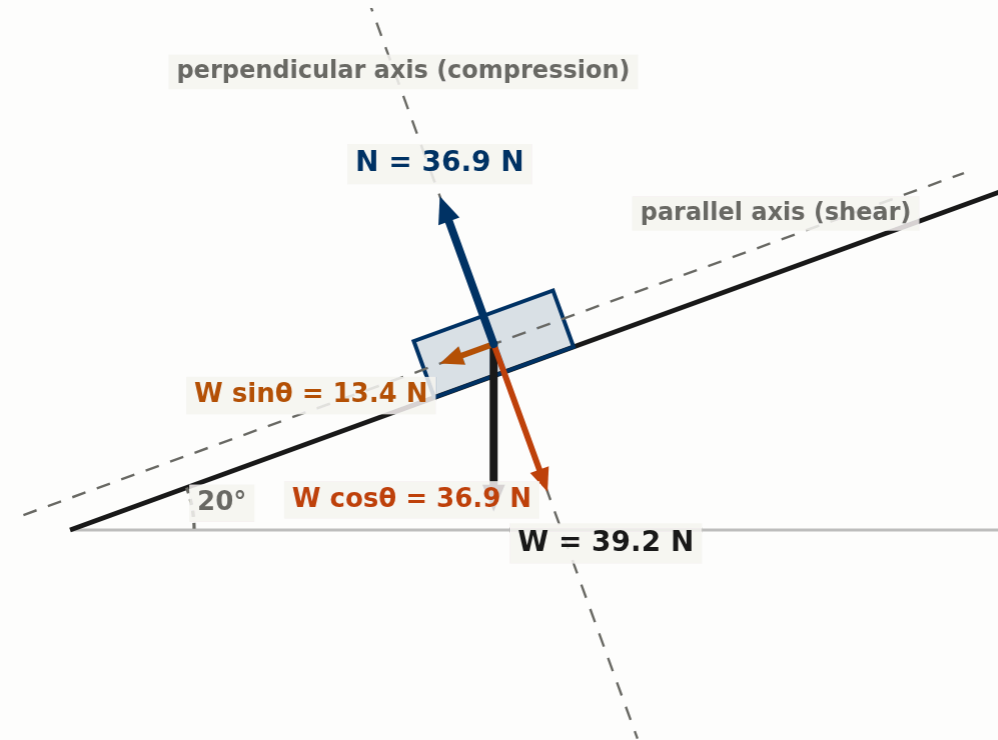
$$F_{gx} = 1000\text{N} \cdot \sin(20^\circ) = 342\text{N}$$



Tilt the axes, not gravity

With contact forces it is customary to build an axis system that is **parallel and perpendicular to the contact surface**. Then every force in the problem is resolved along those two axes.

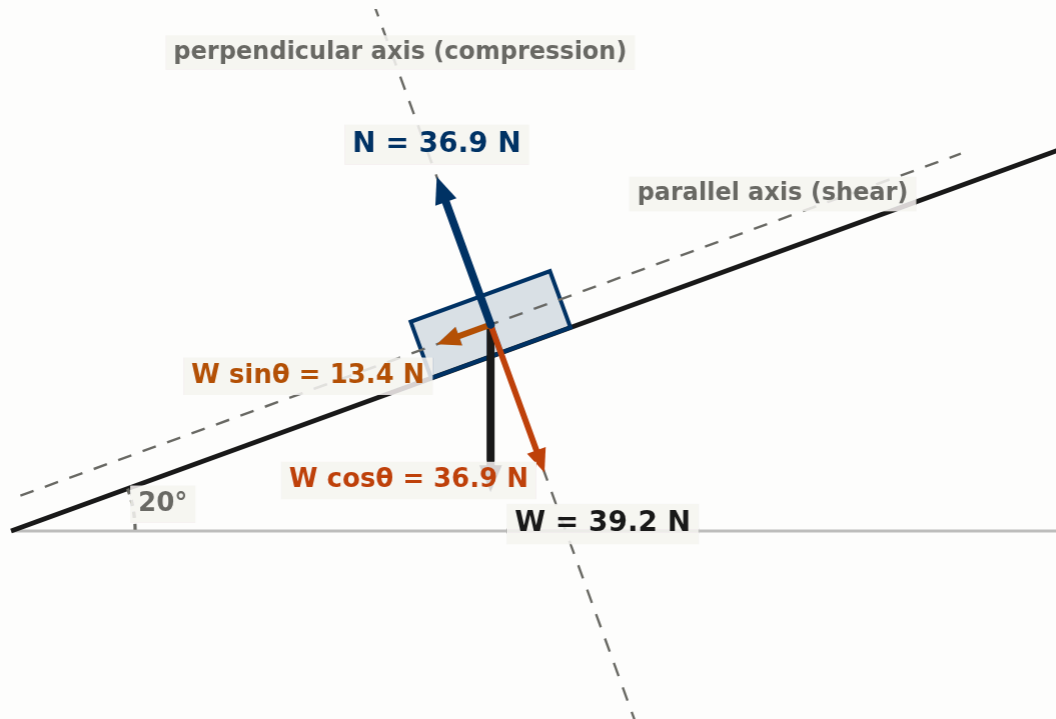
- **Perpendicular** to the surface — the **compression** axis
- **Parallel** to the surface — the **shear** axis
- The normal force is equal and opposite to the component of gravity that is perpendicular to the surface



At 20° the normal force is **36.9 N**, not 39.2 N — the missing 2.4 N has become **13.4 N of shear** down the slope

Worked example — 4 kg on a 20° slope

We need the angle of the slant (20°) and the mass of the object (4 kg). Draw the diagram and the axes, find the weight, then resolve it along the axes.



4.0 kg on a 20° slope

the weight

$$W = mg = 4.0 \times 9.81 = \mathbf{39.24 \text{ N}}$$

perpendicular — pressed into the surface

$$W \cos\theta = 39.24 \cdot \cos 20^\circ = \mathbf{36.87 \text{ N}}$$

parallel — sliding down the surface

$$W \sin\theta = 39.24 \cdot \sin 20^\circ = \mathbf{13.42 \text{ N}}$$

N
36.9 N

shear
13.4 N

N / W
94%

At 20° the normal force is **36.9 N**, not 39.2 N — the missing 2.4 N has become **13.4 N of shear** down the slope

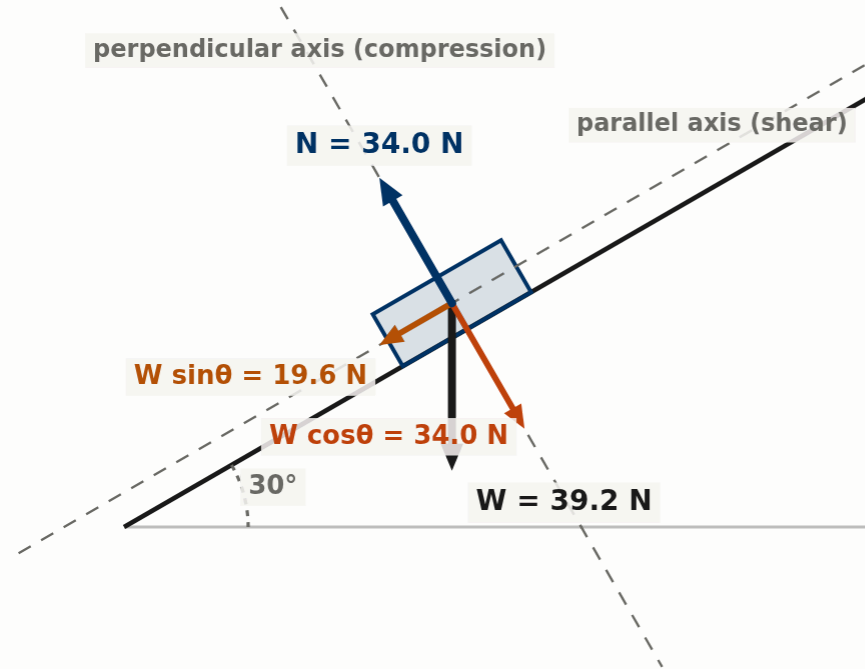
The general rule

For any incline of angle θ , the two components of the weight are always:

$$\text{parallel} = mg \sin \theta$$

$$\text{perpendicular} = mg \cos \theta$$

Where mg is the weight of the object. At $\theta = 0$ the whole weight is perpendicular; at $\theta = 90^\circ$ all of it is shear.



At 30° the normal force is **34.0 N**, not 39.2 N — the missing 5.3 N has become **19.6 N of shear** down the slope

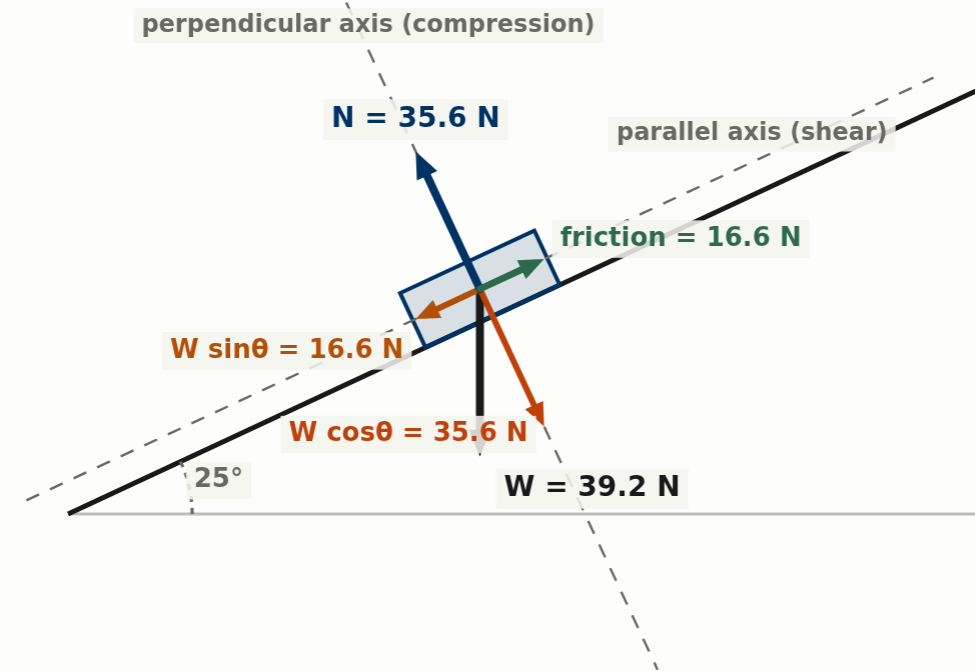
Gravity never changes — it is the same arrow straight down at every angle. What changes is the axis system we measure it against, which is why the normal force falls away as the slope steepens.

Slope angle 30°

The parallel component of gravity tends to pull the block **down the surface**.

Friction force must be **equal and opposite** to the parallel force, or the block would slide.

Friction is proportional to the **normal force** and acts parallel to the surface. More on friction in a following lecture.



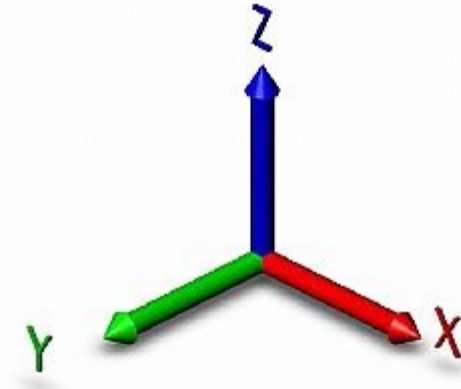
At 25° the normal force is **35.6 N**, not 39.2 N — the missing 3.7 N has become **16.6 N of shear** down the slope

Gravity never changes — it is the same arrow straight down at every angle. What changes is the axis system we measure it against, which is why the normal force falls away as the slope steepens, and why friction has to hold up whatever the shear is.

Forces in 3D

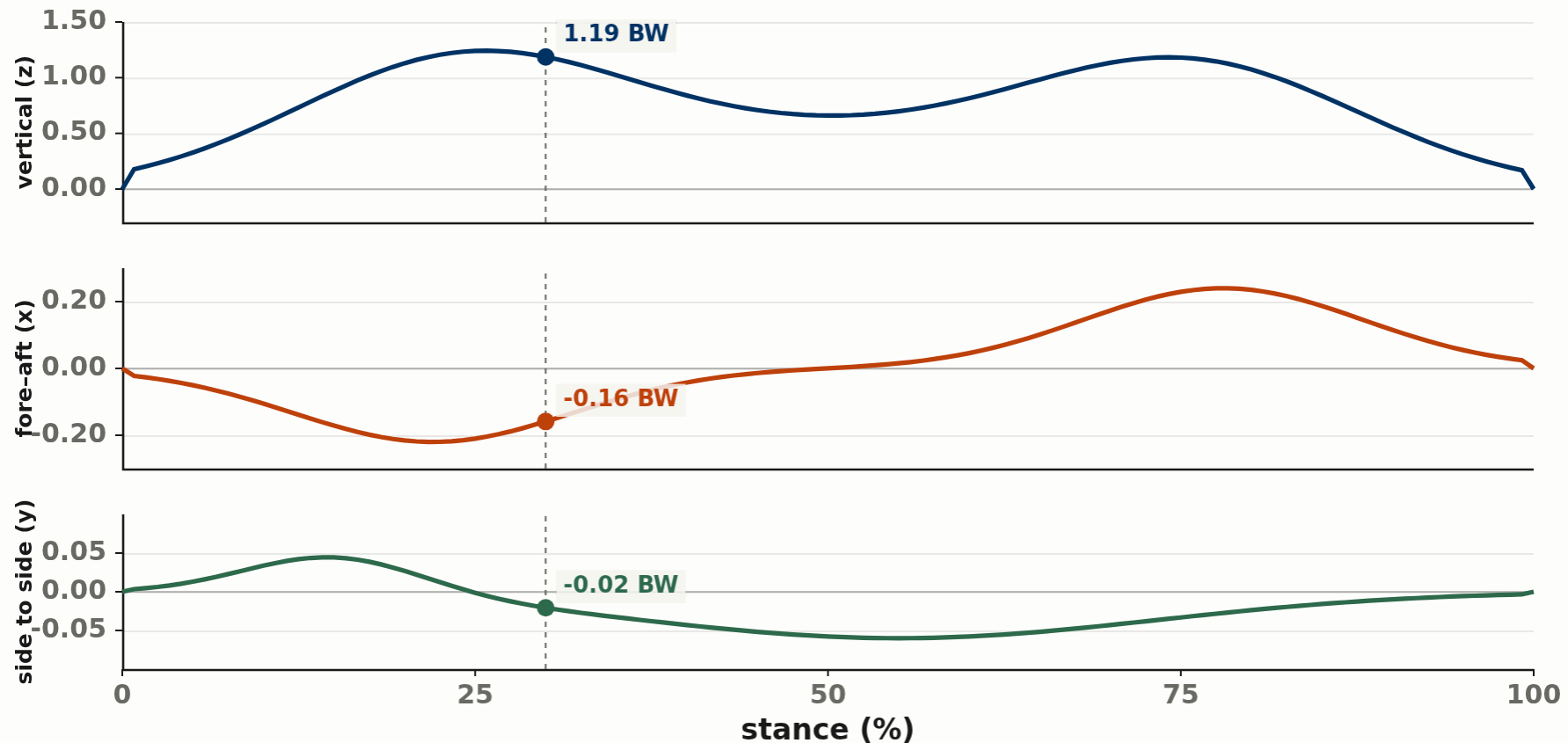
When we move we exert force in **all directions**.

Both internal and external forces are resolved into components with reference to **anatomical planes** and axis systems.



One step, three components

In the sagittal plane, the force we exert on the ground is not just downward — part of it is parallel to the surface. In the frontal plane there is a third, sideways part.



30% of stance · vertical **1.19** · fore-aft **-0.16** · side to side **-0.02** BW

The fore-aft component is **negative**: the ground is pushing backwards on you. This is the braking half of the step. The vertical component is twenty times the others, which is why it is the one everybody quotes — and why the small ones are where the interesting things hide.

Walking, seen from two planes



Measuring external force

- Strain gauges
- Force transducers
- Force plates
- Load cells
- Springs
- Piezoelectric crystals
- Dynamometers

An applied force deforms something, and that deformation is measured and amplified by the electronics. Every device on this list is a way of making a deformation into a voltage.



Single-axis devices

In many clinical settings force transducers are simple **single-axis** devices.

A hand-held dynamometer is a good example, used to determine muscle strength capabilities and weaknesses.

One axis means one number — and it is only the right number if the device is held along the line of the force you meant to measure.



Force plates and instrumented treadmills

The gold standard measurement of force in biomechanics.

PROS

Accurate measure of ground reaction force

Reliable

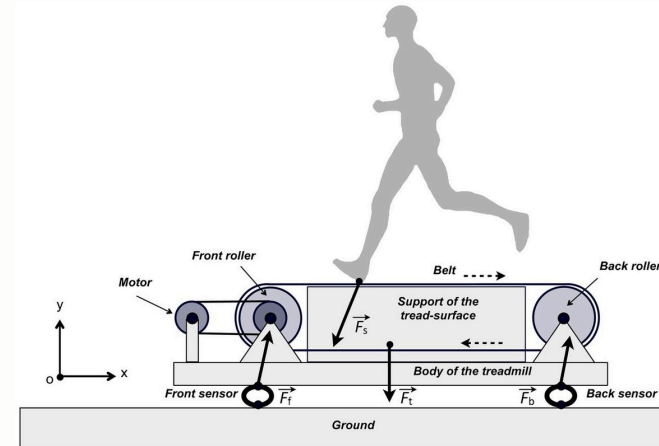
Multidimensional — all three components

CONS

Difficult to use in the field

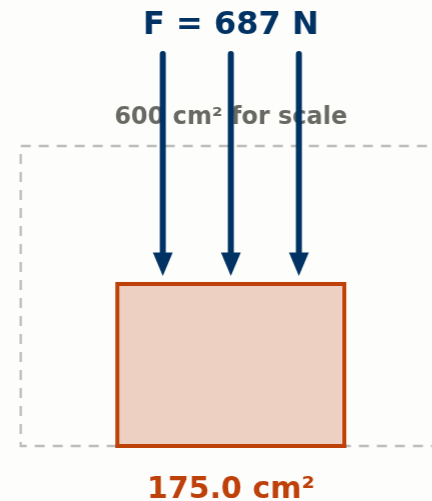
Cannot measure pressure distribution under the foot

One or two contacts per trial



Pressure — the cheap alternative

Often it is not feasible to measure external force directly. Pressure is the **force per unit area** directed perpendicular to a surface: $P = F / A$.



$$P = F / A$$
$$687 \text{ N} / 0.0175 \text{ m}^2$$

39.2 kPa

one foot flat


log scale — the spike is off the end of a linear one

687 N on **175.0 cm²** = **39.2 kPa**

The force never changed — only the area it is spread over. That is why a pressure sensor is not a force sensor: two readings can differ by a hundred times with the same load on the foot.

Body mass



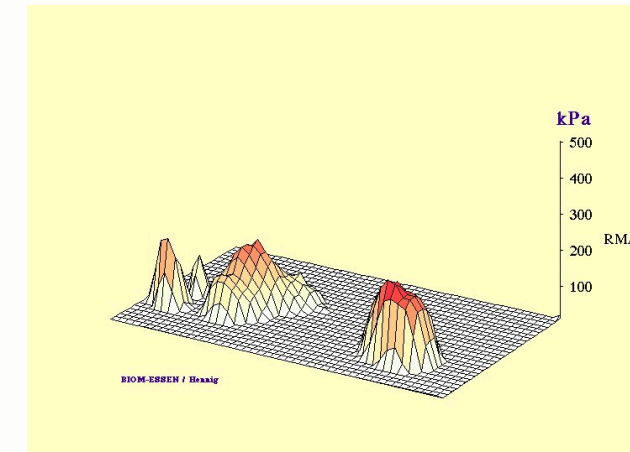
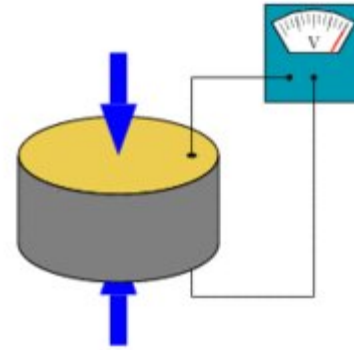
70 kg

How pressure sensors work

Most common pressure sensors use the **piezoresistive effect**.

- *Piezo* — to squeeze or press
- The piezoresistive effect is a change in the electrical resistivity of a semiconductor or metal when mechanical strain is applied

A piezoresistive sensor can be very thin, light and flexible, and many can be tiled together to make a matrix.



Pressure sensing under the foot

PROS

Portable, thin, flexible, lightweight

Approximation of force

Accurate for the *timing* of foot loading

Measures the distribution a force plate cannot

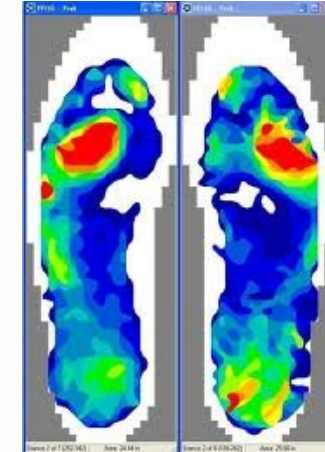
CONS

Unidimensional — vertical loading only

Precision, accuracy and reliability need testing

Calibration drifts with use

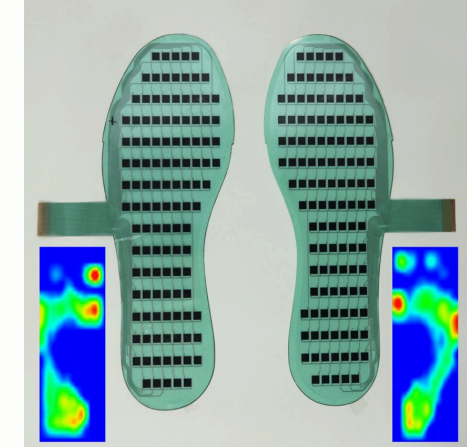
Sensor array is coarse



Insole systems

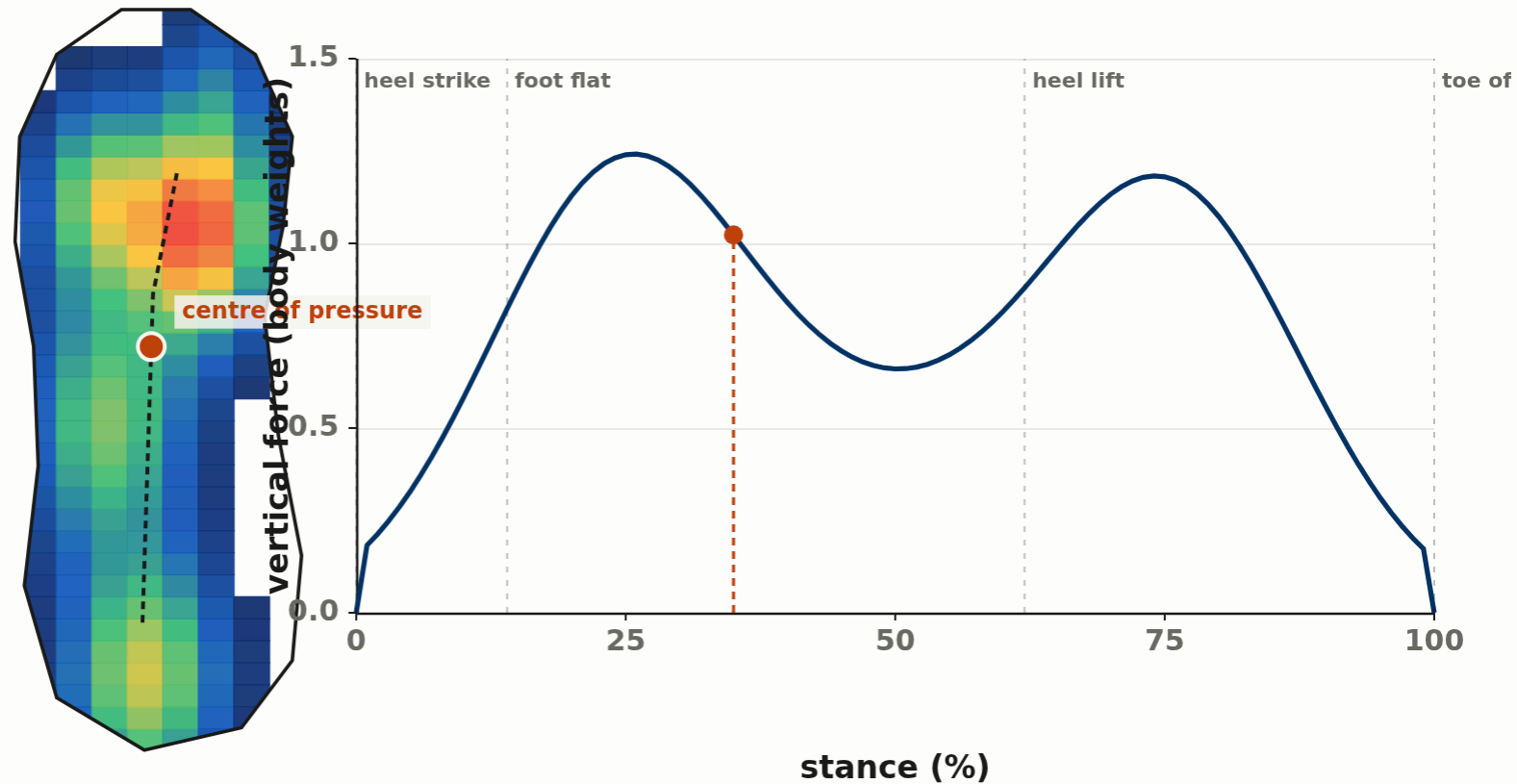
There are many insole **plantar pressure measurement systems** (PPMS). They offer the possibility of measuring external pressure — and calculating an approximate external force — outside the lab.

Research grade, commercial and one-off engineering builds. Even Nike made one. All of them trade accuracy for the ability to walk out of the door.



Centre of pressure

The centre of pressure is the **instantaneous point of application** of the ground reaction force. Watch where it goes during one step.



35% of stance · rolling forward through the midfoot · vertical force **1.02** body weights

The centre of pressure is the single point the ground reaction force acts through at that instant — not where the pressure is highest, but the average of the whole field. It sweeps heel to toe in about 0.6 s of walking.

Insoles against force plates

Instrumented devices

Force and pressure sensors can be added to mobility devices so that the object itself becomes **instrumented**, and we can measure the forces an individual produces during real activity.

Pedals, handrims, crutches, walker handles — anywhere a person pushes on equipment is somewhere a transducer can go.

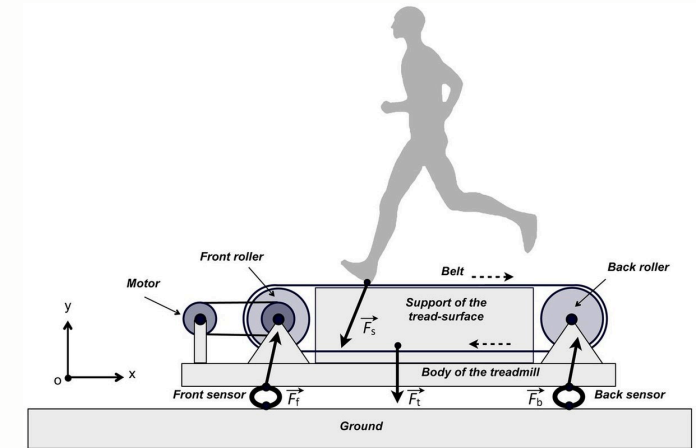


Force plate application — running

Two things get compared:

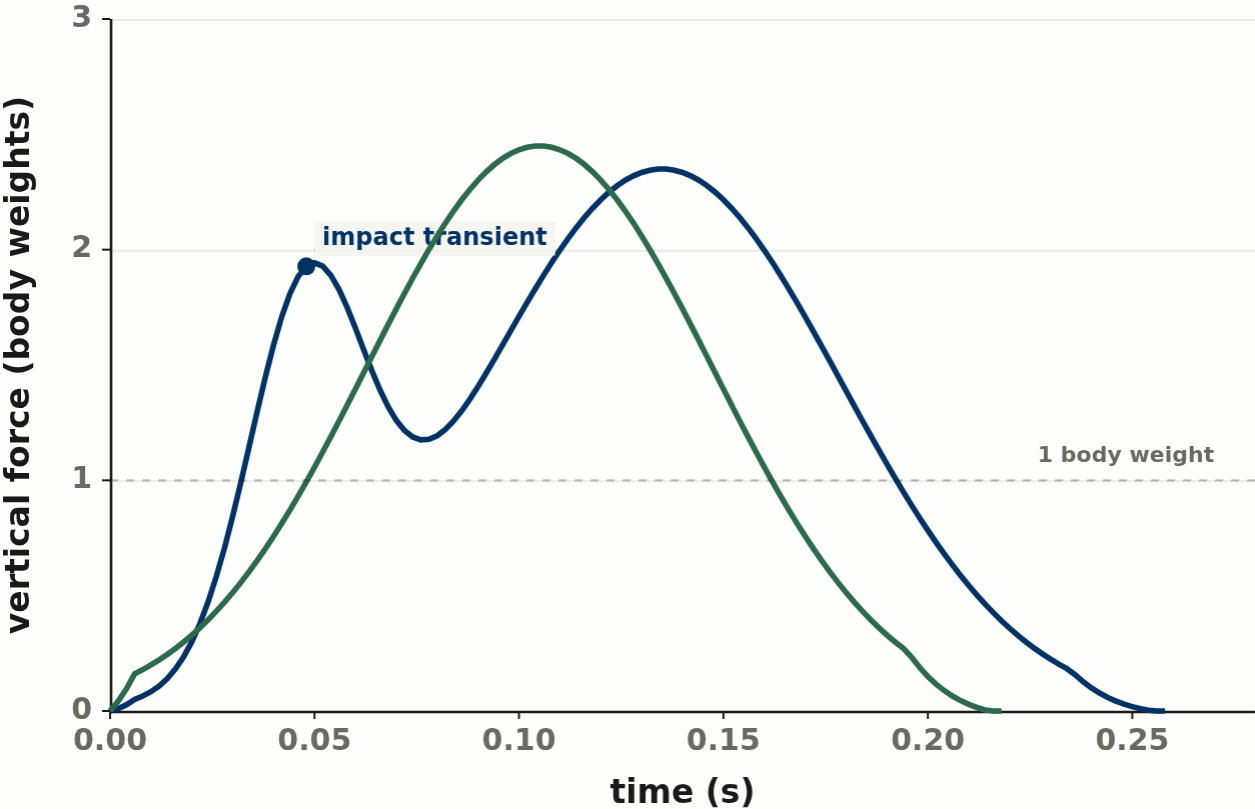
- **Running style** — heel strike or forefoot strike
- **Footwear** — barefoot, shoes, minimal shoes

Lieberman et al., *Foot strike patterns and collision forces in habitually barefoot versus shod runners*, Nature 463, 2010.



What the force trace shows

Every style lands on about the same peak force. What separates them is the **spike in the first 50 milliseconds** — and how fast the load arrives.

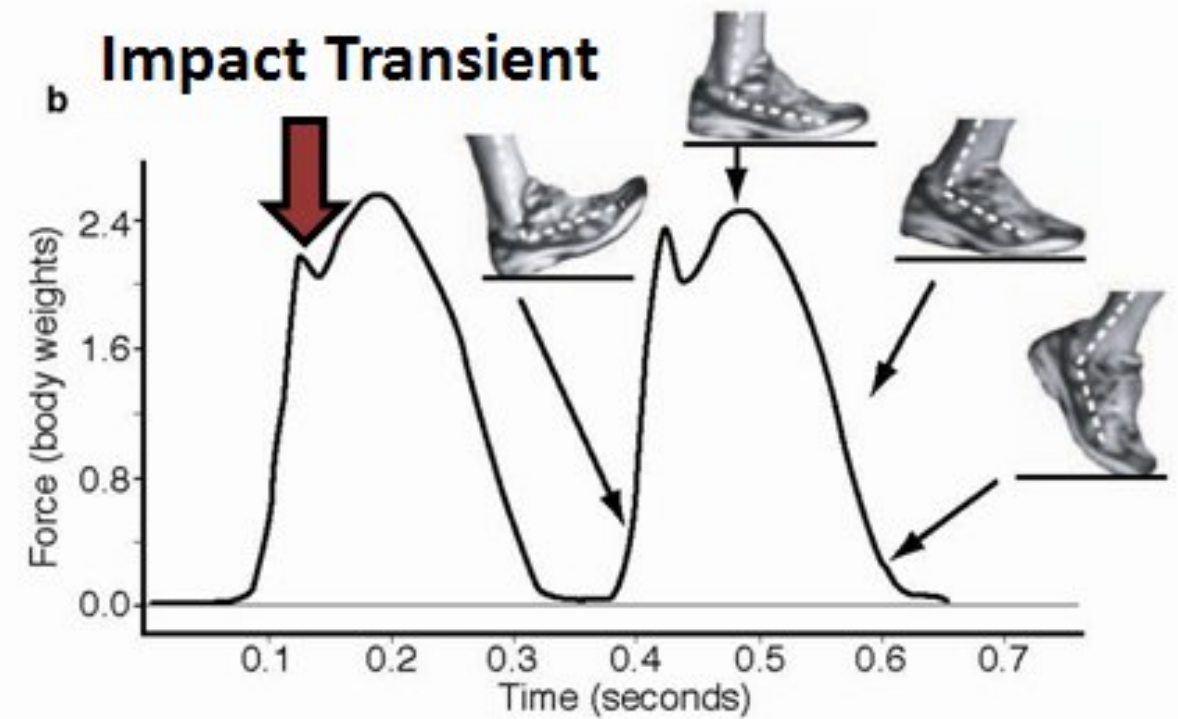
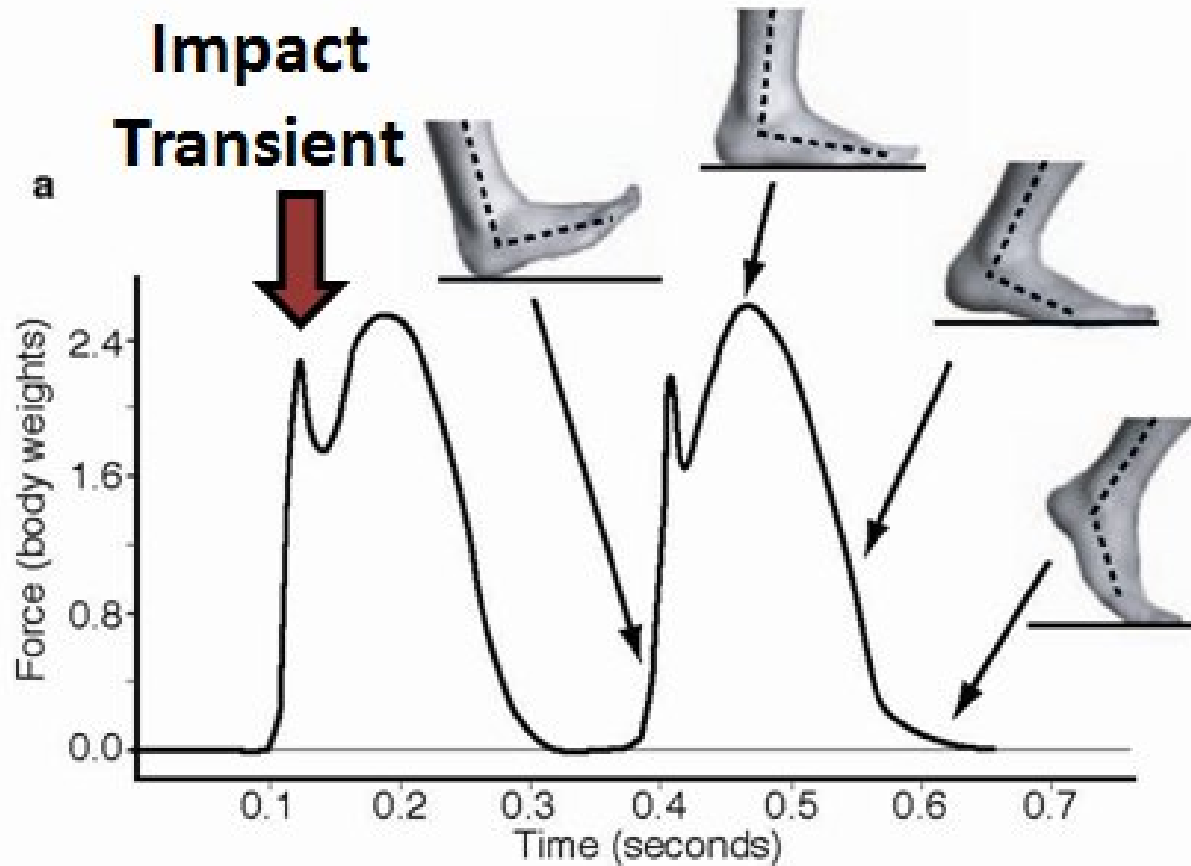


peak force and how fast it arrives

	peak	in N	rate
shod, heel strike	2.35	1614	75
barefoot, heel strike	2.71	1860	264
barefoot, forefoot	2.45	1682	35

Peak in body weights, the same force in newtons for a 70 kg runner, and the steepest rise in body weights per second.

Every style ends up near the same peak. What separates them is the **spike in the first 50 ms** and how quickly the load arrives — which is the part a heel strike in a cushioned shoe softens, and a bare heel does not.



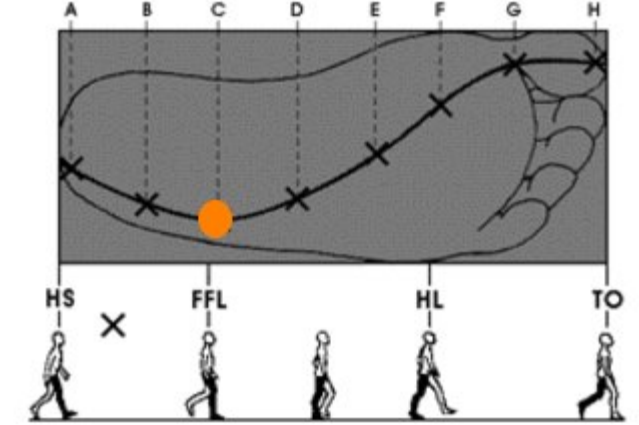
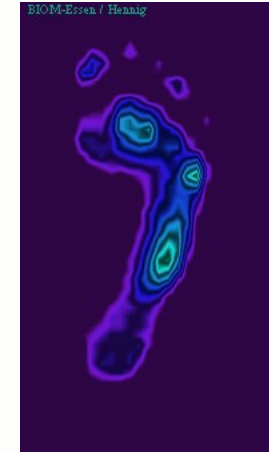
Lieberman et al., *Nature* 463, 531–535 (2010). The arrow marks the impact transient.

Pressure follows the same story

Important changes in the **force profile** and in the **kinematics** can both be linked to what the plantar pressure is doing.

A forefoot strike puts the first load under the metatarsal heads; a heel strike puts it under a heel pad a few centimetres behind the ankle, with the leg almost straight.

Daniel Lieberman's group keeps its material at barefootrunning.fas.harvard.edu

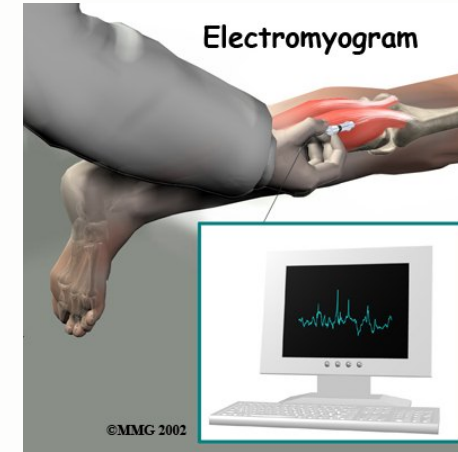
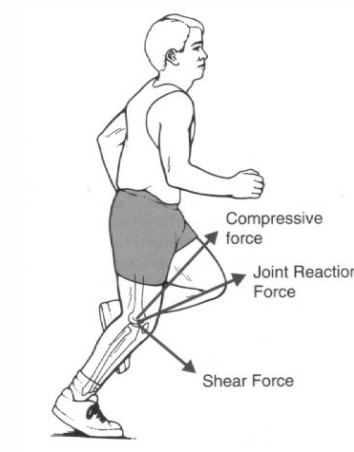


Back to the inside

Force plates and pressure sensors measure **external** forces. It remains difficult to measure the forces at the bones, joints and tendons directly.

But we can **approximate** the internal forces:

- **Inverse dynamics** — work backwards from the measured external force and the measured motion
- **Electromyography** — measure the electrical activity of the muscles that produced them



What to take away

- A force is a **push or a pull** — a vector with a magnitude, a direction, a line of action and a point of application.
- $F = ma$, and weight is just gravity's version: $W = mg$.
- Several forces are combined by **resolving each into components**, adding the components, and rebuilding one resultant. Check the quadrant.
- On a slope, gravity does not tilt — the **axes** do. The perpendicular component is $mg \cos \theta$ and the shear is $mg \sin \theta$.
- External force is measured well by **force plates** and approximated cheaply by **pressure sensors**, which measure distribution and timing rather than force.
- Internal forces are mostly **inferred**, not measured.

- McGinnis, *Biomechanics of Sport and Exercise*, Chapter 1, pp. 20–32 and 39.
- Lieberman, D. E. et al. (2010). Foot strike patterns and collision forces in habitually barefoot versus shod runners. *Nature* **463**, 531–535.
- Dyer, P. S. & Bamberg, S. J. M. Instrumented insole vs. force plate: a comparison of centre of plantar pressure. *IEEE EMBC*.
- Harvard Skeletal Biology Lab — barefootrunning.fas.harvard.edu

The running force traces in this deck are modelled on the published curves — peak, timing and loading rate follow Lieberman et al. They are for illustration, not data.